



INDIAN INSTITUTE OF TECHNOLOGY MADRAS
ZANZIBAR CAMPUS

Mtech Data Science and AI
 School of Science and Engineering
 Z5008: Big Data Laboratory

**Project Proposal: ZECO Revenue Protection & Customer Intelligence Platform:
 Real-Time Analytics for Prepaid Electricity**

Domain: Prepaid Electricity Consumption (ZECO Data)

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1. Problem Statement

Zanzibar Electricity Corporation (ZECO) manages prepaid electricity meters across Unguja and Pemba via a token-based vending system. While ZECO has rich transactional data from vending points, it lacks a real-time analytics platform. This results in: no visibility into consumption patterns, inability to detect tampering/fraud, poor demand forecasting, and no predictive insights for consumers. This project builds a production-grade Big Data platform ingesting ZECO's vending data in real-time, processing it via a Lakehouse architecture, training ML models for anomaly detection and consumption prediction, and serving insights via API and dashboards.

Who benefits

Who Benefits: ZECO Utility companies benefit from reduced revenue leakage; Revenue Assurance teams benefit from real-time "at-risk" flags for technical audits; and Marketing teams benefit from data-driven insights to trigger customer retention offers.

2. Domain and Dataset

Domain: Energy (Prepaid Electricity — ZECO context)

Source: Self-collected Research Dataset (Custom-curated for the Zanzibar context).

Dataset Size: 4.85 GB ($\approx$ 5 Million transaction records).

Availability: Data is fully available and stored locally for immediate ingestion into the Kafka pipeline.

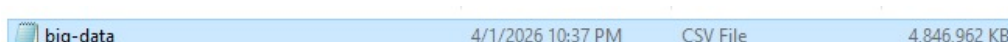


Figure 1: File details for big-data.csv showing 4,846,962 KB size.

Dataset Characteristics

Data satisfies the 5 Vs of Big Data

Volume: Variety: Structured. Data types Streaming:

Data types: Real historical ZECO vending data including: transaction records (meter ID, timestamp, token value, payment amount, vending point) and meter metadata (installation date, customer segment). Data spans 12–24 months with 50,000–100,000 transactions/month.

Streaming: Kafka producer replays historical transactions at accelerated speed (100× real-time) to simulate live streaming.

3. System Architecture

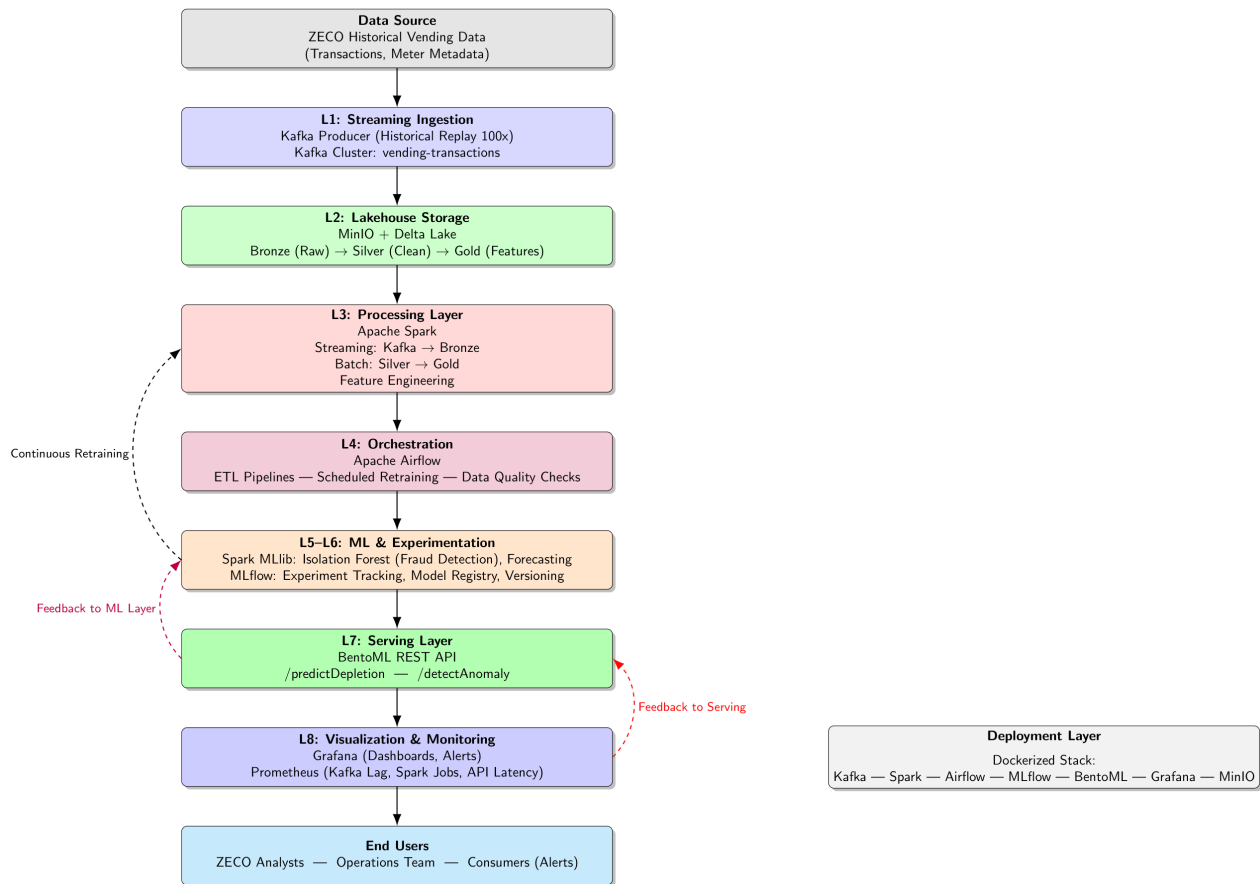


Figure 2: Architecture of the ZECO ML & Monitoring System with feedback loops.

Layer	Technology	Justification
Ingestion	Apache Kafka	Fault-tolerant, scalable replay of historical transactions
Storage	MinIO + Delta Lake	S3-compatible lakehouse with ACID transactions
Processing	Apache Spark	Unified batch + streaming; native Delta Lake integration
Orchestration	Apache Airflow	Scheduled batch jobs, model re-training, reporting
ML Training	Spark MLlib	Direct training on lakehouse data; no data movement
Experiment Tracking	MLflow	Parameter/metric tracking; model registry for versioning
Serving	BentoML	High-performance REST API with native Docker support
Monitoring	Grafana + Prometheus	System metrics (Kafka lag, Spark status) + business metrics

4. Novelty and Uniqueness

1. **Real ZECO Data:** Using actual Zanzibar electricity vending data makes the project locally relevant and impactful.
2. **Historical Replay as Streaming:** Kafka-based replay transforms static records into a live streaming pipeline, demonstrating a migration path from legacy systems.
3. **Anomaly Detection for Revenue Protection:** Isolation Forest model identifies suspicious patterns (unusual depletion intervals, irregular vending volumes, sudden consumption drops).
4. **Token Depletion Prediction:** Time-series forecasting (ARIMA/Prophet) using Spark Pandas UDFs to train models in parallel.
 - **Prophet:** Utilized for individual meter-level forecasting to capture non-linear trends and Zanzibar-specific seasonality/holidays.
 - **ARIMA:** Utilized for aggregate vending station trends where data is more stationary.
5. **Full Lifecycle MLOps:** Models retrained via Airflow, evaluated in MLflow, promoted via registry, and served via versioned API.
6. **Train/Test Split Strategy:** ****Temporal Split:**** A chronological split (80% Training / 20% Testing) will be used. Shuffling is disabled to maintain the time-series integrity and prevent data leakage.

5. Risk Assessment and Mitigation

Risk	Likely	Impact	Mitigation
Data Privacy	Med	High	Anonymize meter IDs; use aggregated data only
Data Format Issues	Med	Med	EDA preprocessing; Spark job for cleaning
Infrastructure Complexity	High	Med	Single <code>docker-compose.yml</code> for all services
Replay Performance	Med	Med	Optimized producer; rate limiting for realistic patterns
Model Performance	Med	Med	Start with simple baselines; regular retraining via Airflow
Team Coordination	Low	Med	GitHub; daily 15-min stand-up

6. Conclusion

This project delivers a production-grade Big Data platform transforming ZECO's vending data into a strategic intelligence asset. By ingesting real historical data, simulating streaming, building models for token depletion prediction and anomaly detection, and serving insights via APIs and dashboards, we provide a replicable blueprint for modernizing Zanzibar's electricity infrastructure while meeting all 8-layer architecture requirements.

Keywords: Big Data, Lakehouse, MLOps, Spark, Kafka, Delta Lake, Zanzibar, Anomaly Detection

Submitted in partial fulfillment of Z5008 Big Data Laboratory requirements.